

Presentation on the History of AI



Alan Turing published his work "Computer Machinery and Intelligence" which eventually became The Turing Test, which experts used to measure computer intelligence

Early Foundations of AI

Foundational period (1940s-1950s)

Early concepts: The idea of thinking machines can be traced to ancient mythology and fictional works, but the modern era began with mathematician Alan Turing.

Theoretical work: In 1935, Turing described a theoretical "universal machine" that could compute anything, and in 1950, his paper "Computing Machinery and Intelligence" introduced the concept of the Turing Test to determine if a machine could exhibit intelligent behavior indistinguishable from a human.

Birth of a field: The Dartmouth workshop in 1956 is considered the official birth of AI as a field, where the term "artificial intelligence" was first used.

AI Winters and expert systems (1960s-1980s)

Early programs: The 1960s and 1970s saw the development of early AI programs that could solve problems and play games, but progress was slow.

AI Winters: The field experienced periods of reduced funding and interest, known as "AI winters," due to a lack of computing power and the limitations of early approaches.

Expert systems: The 1980s saw the rise of expert systems, which were AI programs designed to mimic the decision-making ability of a human expert in a specific field.

Machine learning and modern AI (1990s-Present)

Rise of machine learning: The 1990s saw the emergence of machine learning, where AI systems could learn from data without being explicitly programmed, and neural networks gained traction in the 2000s.

Big Data and GPU acceleration: The 2000s and 2010s saw a major AI boom fueled by an abundance of "big data," more powerful computers, and the use of graphics processing units (GPUs) to accelerate neural network training.

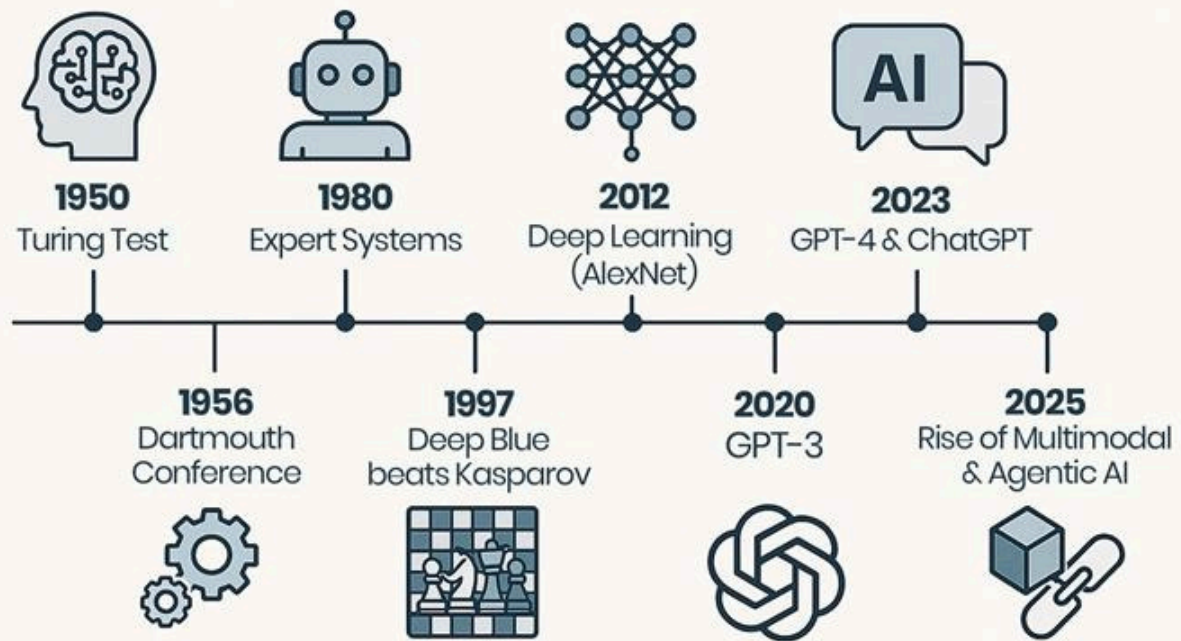
Deep learning breakthroughs: IBM's Deep Blue defeating chess champion Garry Kasparov in 1997 was a major milestone, followed by IBM's Watson winning Jeopardy! in 2011 and advances in image recognition and other areas in the 2010s and 2020s.

Generative AI: The 2020s have seen a focus on generative AI, which can create new content, leading to both excitement and ethical concerns about its impact.

Era Of Agentic AI

The Agentic AI Era: After the Dawn, Here's What to Expect ...The "Era of Agentic AI" refers to the current and future phase of artificial intelligence where systems move beyond just analysis to autonomously take action, reason, plan, and execute tasks to achieve goals.

HISTORY OF ARTIFICIAL INTELLIGENCE



How Large Language Models (LLMs) work 🤖

A Large Language Model (LLM) is an advanced AI system trained on massive datasets to understand, generate, and process human-like language. Using a deep-learning architecture known as a transformer, LLMs can perform complex language tasks with high accuracy and are foundational to many generative AI applications.

How LLMs work:

The core of an LLM's capability is its training process and architecture.

Tokenization and embeddings: The model breaks down text into smaller units called tokens (words, sub-words, or characters). These tokens are then converted into numerical representations called embeddings, which capture their semantic meaning and context.

Fine-tuning: After pre-training, an LLM can be fine-tuned on smaller, more specific datasets. This adapts the model to perform specific tasks, such as medical transcription or generating code, with greater accuracy.

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Major Breakthroughs

Key breakthroughs in large language models (LLMs) have transformed their capabilities through innovations in architecture, scale, and reasoning techniques. These advances have enabled LLMs to progress from generating text to handling complex reasoning, integrating multiple data types, and becoming more useful and reliable for a wide range of applications.

Key architectural and scaling breakthroughs

- **The Transformer architecture (2017):** This was arguably the most significant breakthrough, laying the foundation for all modern LLMs. Introduced in the paper *Attention Is All You Need*, the Transformer model used a self-attention mechanism to process entire sequences of data in parallel, replacing the slower, sequential processing of older recurrent neural networks (RNNs).
- **The large-scale model era (2020):** In 2020, OpenAI released GPT-3, a 175-billion parameter model that demonstrated remarkable abilities in generating coherent and contextually relevant text with minimal or no fine-tuning. Its success showed that increasing model size and data could unlock new capabilities, leading to the "scaling laws" era of LLM development.
- **Mixture of Experts (MoE) architecture (2023):** This architecture improved efficiency by combining several "expert" neural networks. Instead of using the entire model for every task, a gating network selectively routes the input to the most relevant experts, reducing the computational cost for training and inference. Mistral AI's Mixtral 8x7B and OpenAI's suspected GPT-4 architecture utilize this approach.

Train Modern LLMs:

Training modern Large Language Models (LLMs) is a complex, resource-intensive process composed of distinct phases: pre-training, supervised fine-tuning (SFT), and post-training alignment. This multi-stage approach is used by major tech companies and is increasingly accessible to others via open-source models and cloud-based infrastructure.

The modern LLM training pipeline

1. Data collection and preparation

A high-quality, diverse, and massive dataset is the foundation of any powerful LLM. The process typically involves:

Sourcing data: Collecting text spanning trillions of tokens from diverse sources, such as web crawls (like Common Crawl), books, academic papers, and code repositories.

Cleaning and preprocessing: Rigorous cleaning removes irrelevant noise, non-textual elements, and duplicate content to prevent the model from overfitting. Tools like FAISS can perform semantic deduplication at scale.

Mitigating bias: Filtering out harmful, biased, or toxic content is crucial for ethical AI development. Some modern techniques use classifiers trained on human-annotated data to assess readability and quality.

Synthetic data: To augment data for high-value tasks like mathematical reasoning, some modern pipelines generate additional training examples. For instance, the DeepSeek-R1 model used reinforcement learning to create verified, chain-of-thought samples.

Tokenization: The prepared text is converted into numerical tokens that the model can process. Methods like Byte-Pair Encoding (BPE) are used to efficiently represent both common and rare words.

2. Pre-training (Foundation building)

This is the most computationally expensive phase, consuming the vast majority of the training resources.

Self-supervised learning: The model learns language patterns by training on an immense, unlabeled text corpus. A common objective is next-token prediction, where the model learns to guess the most probable next word in a sequence. This establishes the model's foundational understanding of grammar, facts, and semantics.

Architecture: Most modern LLMs are based on the Transformer architecture, which uses a self-attention mechanism to weigh the importance of different tokens in a sequence. This enables the model to understand context and long-range dependencies efficiently.

3. Post-training (Behavioral refinement)

This phase focuses on shaping the pre-trained model to be helpful, safe, and aligned with human instructions and preferences.

Supervised fine-tuning (SFT): The model is trained on a smaller, high-quality dataset of curated examples, consisting of prompt-response pairs. This "instruction tuning" teaches the model to follow directions and respond in a helpful, conversational manner.

Reinforcement learning: This technique further improves the model by incorporating human feedback.

Reinforcement Learning from Human Feedback (RLHF): A separate "reward model" is trained on human rankings of model responses. The LLM is then optimized to produce outputs that the reward model predicts humans would rate highly.

Direct Preference Optimization (DPO): A simpler and more stable alternative to traditional RLHF, DPO directly trains the LLM to prefer chosen responses over rejected ones.

Reasoning and tool use: Advanced techniques use reinforcement learning with verifiable rewards to train models on complex problems (like math or coding). This encourages the model to develop and show a "chain-of-thought" for better problem-solving.

Key considerations and techniques

Computational resources: Training modern LLMs requires immense computational power, typically from thousands of GPUs or TPUs running in parallel.

Distributed training: Techniques like data, model, and pipeline parallelism are used to split the workload across many devices. Libraries like DeepSpeed and Megatron-LM orchestrate this process.

Efficiency: Optimization is critical for managing the cost and time of training.

Parameter-Efficient Fine-Tuning (PEFT): Methods like LoRA (Low-Rank Adaptation) and QLoRA freeze most of the pre-trained model's parameters and train only a small number of new, efficient ones. This drastically reduces the resources needed for fine-tuning.

Mixed-precision training: This technique uses lower-precision formats like FP8 or BF16 for some calculations to reduce memory usage and speed up training.

Evaluation: Throughout the process, models are continuously evaluated using benchmarks and human feedback to measure performance and address issues like bias or "hallucinations".

